

HOMOLOGICAL METHODS IN ALGEBRAIC GEOMETRY

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1. INTRODUCTION

Algebraic geometers study geometric objects by translating geometric questions into algebra. Schemes model spaces cut out by polynomial equations, while their sheaves and cohomology encode global information. Derived categories, rooted in homological algebra and category theory, organize complexes of sheaves and their cohomology. For a scheme or algebraic space X , the derived category $D_{\text{qc}}(X)$ of complexes with quasi-coherent cohomology packages a substantial amount of the homological information attached to X .

Extracting geometric information from $D_{\text{qc}}(X)$ and its subcategories is nevertheless difficult. Several structures and categorical invariants have been developed for this purpose. This includes semiorthogonal decompositions [1] or Bridgeland stability conditions [10].

My previous research has studied how geometric and singularity-theoretic information is encoded in derived categories of schemes, algebraic spaces, and algebraic stacks. I developed and applied techniques involving t -structures, approximation, and generation for triangulated categories to the study structure of these derived categories. This work has led to classification results, bounds on invariants like Rouquier dimension, and categorical characterizations of singularities.

Currently, I have two ongoing research programs that lie at the intersection of geometry, homological algebra, K -theory, and singularity theory: *Fourier–Mukai partnership and higher singularities*. These are discussed in the following sections, and together advance our understanding for derived categories of singular spaces.

2. FOURIER–MUKAI PARTNERSHIP

A current focus of mine is on integral transforms (a.k.a. Fourier–Mukai transforms), which were pioneered by Mukai [36]. The terminology comes from classical analysis. Roughly speaking, given projections $p_i: Y_1 \times_S Y_2 \rightarrow Y_i$ and a kernel K on $Y_1 \times_S Y_2$, the corresponding integral transform is

$$\Phi_K(E) := \mathbf{R}(p_2)_*(\mathbf{L}p_1^*E \otimes^{\mathbf{L}} K).$$

This mirrors the construction of an analytic integral transform: pull a function to a product, multiply it by a kernel, and integrate out one variable. The classical Fourier transform corresponds to $e^{2\pi ixy}$.

The relevance of integral transforms, and hence derived categories, in birational geometry is well known [9, 24, 7]. In many settings, birational operations such as flips, flops, and blow ups are reflected by equivalences or semiorthogonal decompositions of derived categories. Many of the resulting functors can be realized as integral transforms. Suitable kernels may induce derived equivalences, allowing a complicated variety to be studied through a Fourier–Mukai partner with potentially more tractable homological information. Such partners often arise as moduli spaces of coherent sheaves or complexes [5].

2.1. Central problem. Integral transforms are especially natural in families: one can ask how an equivalence varies from fiber to fiber and what it preserves when singular fibers occur. This leads to the motivating question:

Question 1. *Is the loci of points over which a family of integral transforms is fully faithful or an equivalence open?*

A major obstacle for studying Question 1 in the presence of singularities is that existing techniques for studying integral transforms typically require the kernel to be perfect. See e.g. [21, Theorem 5.8] and [6, Proposition 4.9]. In particular, the perfectness hypothesis is essential in existing adjunction techniques: in [2], it is used to represent the unit and counit by morphisms of kernels. Consequently, those arguments do not extend directly to the nonperfect kernels.

This limitation already appears in the study of base change for integral transforms. A classical result of Orlov shows that an integral transform inducing an equivalence between smooth projective varieties remains an equivalence after extension of the ground field [39, Lemma 2.12]. Recently, [20] established an analog for singular varieties.

However, the proofs are considerably different. Orlov’s proof relies on morphisms between perfect complexes representing the unit and counit of the relevant adjunctions, whereas such morphisms need not exist for nonperfect kernels. Consequently, the techniques developed for perfect kernels do not extend to the singular setting.

The following illustrates that nonperfect kernels are unavoidable.

Example 1. Let $k := \mathbb{F}_3(t)$ where t is transcendental. Consider the projective plane curve $f: C \rightarrow \operatorname{Spec}(k)$ given by the equation $y^2z + x^3 - tz^3 = 0$. This is a regular Noetherian scheme as it is normal and of Krull dimension one. However, after base change of C along the field extension $\ell := \mathbb{F}_3(t^{\frac{1}{3}})$, $C \times_k \operatorname{Spec}(\ell)$ is a singular projective curve [26, Remark 16]. Denote by $t': C \times_k C \times_k \operatorname{Spec}(\ell) \rightarrow C \times_k C$ the natural base change morphism. Choose any $K \in D_{\operatorname{coh}}^b(C \times_k C)$ for which its associated integral transform $\Phi_K: D_{\operatorname{qc}}(C) \rightarrow D_{\operatorname{qc}}(C)$ is an equivalence. Then K is not perfect. See [19] for details.

Example 1 shows that any theory of Fourier–Mukai transforms capable of addressing Question 1 must allow nonperfect kernels of integral transforms. Indeed, it shows that integral transforms inducing autoequivalences even on a regular curve do not arise from perfect kernels. This is counterintuitive: the fiber product of two smooth curves is smooth, and so $D_{\operatorname{coh}}^b = \operatorname{Perf}$ for such fiber products. Since C is regular but not smooth, this is not the case, and so $D_{\operatorname{coh}}^b(C \times_k C) \neq \operatorname{Perf}(C \times_k C)$. Thus, even autoequivalences of regular curves may require nonperfect kernels.

2.2. State of the art. In recent joint work, I have developed the corresponding theory for Noetherian algebraic spaces. This completed foundation establishes preservation criteria, base change compatible adjunctions, fiberwise detection, openness of the Fourier–Mukai loci, and autoequivalences of elliptic fibrations:

With Guisado Villalgorido, a PhD student whom I coadvise at BCAM, we prove:

Theorem 1 (Guisado Villalgorido–Lank). *Consider proper flat surjective morphisms $f_1: Y_1 \rightarrow S$ and $f_2: Y_2 \rightarrow S$ of Noetherian algebraic spaces. For any $K \in D_{\operatorname{coh}}^b(Y_1 \times_S Y_2)$ which is relatively perfect over each Y_i , the points $s \in |S|$ where the fiber of Φ_K is fully faithful (resp. an equivalence) is an open subset of $|S|$.*

In other words, Theorem 1 establishes an openness property for the loci of points where fully faithfulness or equivalences occur for base changed integral transforms. Even for schemes, this is quite new, particularly because it applies in arbitrary characteristic.

Our work identifies the complement of these loci with the image of the support of the cones of the unit or counit of an adjunction applied to compact generators. Specifically, fully faithfulness is controlled by the support of the cone of the unit on a compact generator, whereas equivalence is controlled by the union of the supports of the cones of the unit and counit for compact generators.

Neither this description of the loci nor the resulting openness theorem appears to be known even for schemes. This generalizes the algebraic space version of [21, Corollary 5.9] in two ways: first, we do not require perfect kernels; second, the arguments differ since that of [2] are not available. In fact, the techniques here are independent.

2.3. Beyond the state of the art. Existing fiberwise criteria are available principally for schemes or perfect kernels. The completed foundation goes beyond this state of the art by treating Noetherian algebraic spaces, allowing pseudocoherent kernels that need not be perfect, and establishing openness of the associated loci. My goal is to determine which parts of this theory survive after passage to singularity categories.

With my PhD student Guisado Villalgorido, we aim to address:

Question 2. *Under what hypotheses can an integral transform on singularity categories be fully faithful or an equivalence, and when can these properties be detected fiberwise?*

This research program builds naturally on my recent work on integral transforms, relative Fourier–Mukai theory, and categorical methods [17, 20, 33, 4].

2.4. Singularity categories. For a Noetherian algebraic space X , its singularity category is the Verdier quotient $D_{\operatorname{sg}}(X) := D_{\operatorname{coh}}^b(X)/\operatorname{Perf}(X)$. This originates in Buchweitz’s work in the affine setting [11] and was later developed for schemes by Orlov [40].

Derived categories of singular Noetherian spaces are generally less understood than those of smooth spaces. The singularity category isolates why: $D_{\operatorname{sg}}(X) = 0$ if, and only if, X is regular. Its K -theoretic properties have been studied for schemes [41], as have equivalences between singularity categories [23, 25, 35]. In several important examples, such equivalences are induced by integral transforms on D_{coh}^b .

These observations motivate the following objectives, which investigate how the relative Fourier–Mukai theory developed in [19] behaves after passage to singularity categories.

Objective A. 1. *Establish openness of the relatively perfect loci of a kernel.*

Objective A. 2. *Characterize when an induced integral transform on D_{sg} is fully faithful or an equivalence in terms of perfectness of cones of the unit/counit in adjunctions.*

Objective A. 3. *Identify situations which fully faithfulness and equivalence on D_{sg} can be detected fiberwise, and determine the obstructions when these hypotheses fail.*

For a kernel K on $Y_1 \times_S Y_2$, let $\text{RP}_i(K)$ denote the set of points $p \in |S|$ for which the fiber kernel K_p is relatively perfect over $Y_i \times_S \text{Spec}(k)$ with $t: \text{Spec}(k) \rightarrow S$ a representative of p . In [Objective A. 1](#), we will prove that these loci are open. In particular, the work in [\[19\]](#) gives independence of the chosen representative. Preliminary work identifies the support-theoretic mechanism underlying this result in key cases. At every point in $\text{RP}_1(K) \cap \text{RP}_2(K)$, the corresponding fiber integral transform Φ_{K_p} preserves both bounded coherent and perfect complexes. Thus, the intersection of the $\text{RP}_i(K)$ records precisely where fiber transforms induce an exact functor on singularity categories.

For [Objective A. 2](#), suppose that Φ_K admits an adjoint $\Phi_{K'}$. Fully faithfulness of the induced functor on D_{sg} is controlled by whether the cone of the unit $1 \rightarrow \Phi_{K'} \circ \Phi_K$ is perfect for all bounded pseudocoherent complexes, while equivalence also requires similar conditions for the corresponding counit. Preliminary work identifies the failure of perfection for the cones of the unit and counit as the relevant obstruction. This transfers a difficult computation in D_{sg} to a perfectness problem in D_{coh}^b , where the methods of [\[19\]](#) apply.

Finally, [Objective A. 3](#) studies the extent to which these properties are fiberwise. Unlike the situation for D_{coh}^b , preliminary examples indicate that full faithfulness and equivalence on D_{sg} need not be invariant under changing the representatives of a closed point. In fact, projective curves in positive characteristic provide examples in which ascent or descent fails. However, we can show if for every closed point, there exist some representative for which the base changed integral transform is an equivalence or fully faithful, then fiberwise detection occurs. This contrasts the case of [\[19\]](#) where independence of the representative occurred. These results will determine the limits of relative Fourier–Mukai theory on singularity categories.

3. HIGHER SINGULARITIES

My other research program uses mixed Hodge modules to extract information from D_{coh}^b . Saito’s theory of mixed Hodge modules provides a bridge between topology, Hodge theory, and algebraic geometry [\[42, 43\]](#). It is centered on the bounded derived category of mixed Hodge modules $D^b(\text{MHM}(X))$ on an embeddable complex variety X .

For each $p \in \mathbb{Z}$, there is an exact graded de Rham functor,

$$\text{gr}_p^F \text{DR}_X: D^b(\text{MHM}(X)) \rightarrow D_{\text{coh}}^b(X).$$

These functors produce important bounded coherent complexes on X . In particular, the *trivial Hodge module* gives the p -th Du Bois complex,

$$\underline{\Omega}_X^p \cong \text{gr}_{-p}^F \text{DR}_X(\mathbb{Q}_X^H[\dim X])[p - \dim X].$$

Du Bois complexes now play a central role in singularity theory [\[37, 22, 38, 18, 46, 29\]](#).

We say that X has *pre- m -Du Bois singularities* if $\mathcal{H}^j(\underline{\Omega}_X^p) = 0$ for all $0 \leq p \leq m$ and $j > 0$. Examples include toric varieties and various varieties with local complete intersection singularities. Parallel constructions using intersection Du Bois complexes describe refinements of rational singularities.

3.1. State of the art. Recent work has connected higher singularities with algebraic K -theory: Shen related higher Du Bois singularities to K -regularity, a measure of $\mathbb{A}^1$ -invariance of algebraic K -groups [\[45\]](#), while Burke-Popa-Shen study negative K -groups through mixed Hodge theory and higher singularities [\[12\]](#).

The p -th graded de Rham functors induce homomorphisms on Grothendieck groups,

$$K_0(\text{gr}_p^F \text{DR}_X): K_0(\text{MHM}(X)) \rightarrow K_0(\text{coh}(X)).$$

No general result shows that the essential image of a fixed graded de Rham functor generates $D_{\text{coh}}^b(X)$, or that the induced morphism on K_0 is surjective. More broadly, no general categorical description of this image is available. It remains unclear what information is encoded by this morphism on K_0 .

This raises the central question of the research program:

Question 3. *To what extent do the graded de Rham images of mixed Hodge modules generate $D_{\text{coh}}^b(X)$, and can the number of extensions required measure the singularities of X ?*

3.2. Beyond the state of the art. In joint work with Venkatesh, I will address [Question 3](#) using geometric methods and generation techniques à la Bondal–Van den Bergh [8]. Standard generation permits shifts, cones, finite coproducts, and direct summands. We seek a stronger dévissage using only shifts and cones (i.e. no summands). This distinction is essential for K -theory. Indeed, because the defining relations in K_0 arise from distinguished triangles, such a dévissage would imply that the class of every coherent sheaf in $K_0(\mathrm{coh}(X))$ lies in the image of a class arising from a mixed Hodge module.

We have three objectives:

Objective B. 1 (Hodge-theoretic dévissage). *Establish that every object of $D_{\mathrm{coh}}^b(X)$ is generated, using only cones and shifts, by the complexes $\mathbf{R}i_*\underline{\Omega}_Z^0$ where $i: Z \rightarrow X$ ranges over integral closed subvarieties. Deduce that the morphism*

$$K_0(\mathrm{gr}_p^F \mathrm{DR}_X): K_0(\mathrm{MHM}(X)) \rightarrow K_0(\mathrm{coh}(X))$$

is surjective for all $p \in \mathbb{Z}$.

Objective B. 2 (Categorical invariants). *Introduce categorical numerics arising from this dévissage and characterize the higher Du Bois singularities detected by their extremal values.*

Objective B. 3 (Bounds). *Establish bounds for fiber products of varieties.*

This research program builds on my prior research with Hodge-theoretic techniques for higher singularities [27], categorical characterizations of singularities [31, 34, 14, 4], and invariants of derived categories for singular varieties [15, 32, 30].

Throughout, a *variety* means a separated reduced (not necessarily irreducible) scheme of finite type over $\mathbb{C}$. Moreover, a variety is *embeddable* if it admits a closed immersion into a smooth variety.

3.3. Hodge-theoretic dévissage. The graded de Rham functors commute with derived pushforward along proper morphisms. More precisely, for every proper morphism $f: Y \rightarrow X$ of varieties, there exists a commutative diagram:

$$\begin{array}{ccc} D^b(\mathrm{MHM}(Y)) & \xrightarrow{\mathrm{gr}_p^F \mathrm{DR}_Y} & D_{\mathrm{coh}}^b(Y) \\ f_*^{\mathrm{MHM}} \downarrow & & \downarrow \mathbf{R}f_* \\ D^b(\mathrm{MHM}(X)) & \xrightarrow{\mathrm{gr}_p^F \mathrm{DR}_X} & D_{\mathrm{coh}}^b(X). \end{array}$$

Objective B. 1 seeks to establish the following strong form of dévissage: *every $E \in D_{\mathrm{coh}}^b(X)$ can be obtained, using only cones and shifts, from the 0-th Du Bois complexes $\mathbf{R}i_*\underline{\Omega}_Z^0$ obtained by pushing forward the zeroth Du Bois complexes of integral closed subvarieties $i: Z \rightarrow X$. Crucially, direct summands are not used. This places the construction in the setting detected by Grothendieck groups. Our preliminary work provides a tentative proof of this strong dévissage, which is discussed below.*

Thomason’s criterion [47] translates this dévissage into surjectivity on Grothendieck groups:

Goal 1 (Lank–Venkatesh). *Let X be an embeddable variety. For any $p \in \mathbb{Z}$, the induced morphism*

$$K_0(\mathrm{gr}_p^F \mathrm{DR}_X): K_0(\mathrm{MHM}(X)) \rightarrow K_0(\mathrm{coh}(X))$$

is surjective.

This expresses $K_0(\mathrm{coh}(X))$ entirely in terms of classes coming from mixed Hodge modules. Notably, it applies to varieties with arbitrary singularities. Our preliminary proof reduces to $p = 0$ using Tate twists and then proceeds by induction on the Krull dimension. The induction has two key ingredients. First, on the smooth locus $U \subseteq X$, one has $\underline{\Omega}_U^0 \cong \mathcal{O}_U$. Hence, the cone of the natural morphism $\mathcal{O}_X \rightarrow \underline{\Omega}_X^0$ is supported on a proper closed subset. Second, the graded de Rham functors commute with derived pushforwards of proper morphisms. Together, these properties allow for induction on the dimension of the support.

A further input is a coherent dévissage result: the complexes $\mathbf{R}i_*\mathcal{O}_Z$ where $i: Z \rightarrow X$ ranges over all integral subvarieties, generate $D_{\mathrm{coh}}^b(X)$ using only shifts and cones. The available quasi-projective argument uses an ample line bundle but this does not apply to every embeddable variety. Preliminary work develops the required extension in the broader setting of Noetherian algebraic spaces. Although **Objective B. 1** concerns complex varieties, this stronger result is of independent interest.

3.4. Categorical numerics. Motivated by this rigid form of generation, we introduce numerics that measure how efficiently objects can be built from mixed Hodge-theoretic data. For an embeddable variety X and $p \in \mathbb{Z}$, set Ψ_X^p to be the collection in $D_{\text{coh}}^b(X)$ containing the zero object and those of the form $\mathbf{R}i_*\underline{\Omega}_Z^p$ where $i: Z \rightarrow X$ is a closed immersion from integral subvarieties.

For $\mathcal{T}$ a triangulated category and $\mathcal{U}, \mathcal{S} \subseteq \mathcal{T}$, define $\text{lex}^{\mathcal{S}}(\mathcal{U})$ as the smallest $n \geq 0$ such that all objects of $\mathcal{U}$ can be built as an n -fold extension of shifted objects of $\mathcal{S}$. If no such n exists, we set $\text{lex}^{\mathcal{S}}(\mathcal{U}) = \infty$. By convention, lex being zero consists only of the zero object, while lex being one consists of shifts of objects of $\mathcal{S}$.

This is related to ‘level’ introduced in [3] but we do not allow direct summands or finite coproducts. Preliminary work has studied basic properties of lex . Our plan is to study values such as $\text{lex}^{\Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$ and $\text{lex}^{\Psi_X^0 \oplus \Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$. In fact, the preliminary work on [Goal 1](#) implies $\text{lex}^{\Psi_X^0 \oplus \Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p))$ is always finite. Here, $\Psi_X^0 \oplus \Psi_X^p$ means $\{A \oplus B : A \in \Psi_X^0, B \in \Psi_X^p\}$.

The goal of [Objective B. 2](#) is to show that extremal cases recover various classes of singularities.

Goal 2 (Lank–Venkatesh). *Let X be an embeddable integral variety and $m \geq 0$.*

- (1) *X has pre- m -Du Bois singularities if, and only if, for all $0 \leq p \leq m$, one has $\text{lex}^{\Psi_X^p}(\mathcal{H}^0(\underline{\Omega}_X^p)) = 1$.*
- (2) *X has Du Bois singularities if, and only if, $\text{lex}^{\Psi_X^0}(\mathcal{O}_X) = 1$.*

Thus, these numerics measure the failure of a variety to have higher Du Bois singularities. Other numerics appearing in [31, 4, 34, 13] use direct summands. In contrast, the numerics here are defined *without* direct summands or finite coproducts.

Preliminary work provides distinct proof strategies for the two parts of [Goal 2](#). The pre- m -Du Bois argument does not use a splitting criterion, unlike most existing criteria for these singularities. These include [28, Theorem 1], [44, Theorem 4.3], [27, Theorem 3.3], [29, Theorem 10.4], and [48, Theorem F].

By contrast, the Du Bois characterization in [Goal 2](#) currently appears to require a splitting result [44]. Since the two arguments are independent, this dependence does not affect the first part of [Goal 2](#). An additional objective is to determine whether the splitting input in the second part of [Goal 2](#) can be removed.

3.5. Bounds. Next we look towards producing bounds for fiber products. In particular, a crucial step is to obtain sufficient control over the p -th Du Bois complexes of fiber products. Preliminary work with Dirks has produced the following Künneth decomposition, which serves as initial tool here. Let X and Y be embeddable varieties. Denote by f_X, f_Y the projections from $X \times_{\mathbb{C}} Y$ to X and Y . For any $0 \leq p \leq \dim X + \dim Y$, there is a decomposition $\underline{\Omega}_{X \times_{\mathbb{C}} Y}^p \cong \bigoplus_{s+t=p} \mathbf{L}f_X^* \underline{\Omega}_X^s \otimes^{\mathbf{L}} \mathbf{L}f_Y^* \underline{\Omega}_Y^t$.

The decomposition provides the main input for product bounds. It also implies that the fiber product of two pre- m -Du Bois varieties has pre- m -Du Bois singularities. In particular, this generalizes [16, Theorem 3.9] to the setting of higher singularities. Therefore, it naturally leads to the problem of measuring the failure of $X \times_{\mathbb{C}} Y$ to have such singularities in terms of numerics associated with X and Y .

Preliminary arguments suggest we have a proof of the following:

Goal 3. *Let X and Y be embeddable integral varieties. Denote by f_X, f_Y the projections from $X \times_{\mathbb{C}} Y$ to X and Y . Then there are inequalities:*

- (1) $\text{lex}^{\Psi_{X \times_{\mathbb{C}} Y}^0}(\mathcal{O}_{X \times_{\mathbb{C}} Y}) \leq \text{lex}^{\Psi_X^0}(\mathcal{O}_X) \cdot \text{lex}^{\Psi_Y^0}(\mathcal{O}_Y)$
- (2) $\text{lex}^{\Psi_{X \times_{\mathbb{C}} Y}^0}(\mathcal{H}^0(\underline{\Omega}_{X \times_{\mathbb{C}} Y}^0)) \leq \text{lex}^{\Psi_X^0}(\mathcal{H}^0(\underline{\Omega}_X^0)) \cdot \text{lex}^{\Psi_Y^0}(\mathcal{H}^0(\underline{\Omega}_Y^0)).$

4. LONG-TERM DEVELOPMENTS

Over the few years, I will continue to develop these two programs in parallel. The first stage will complete the relative theory of integral transforms on singularity categories and determine the precise scope of fiberwise detection. In the second program, I will extend the Hodge-theoretic numerics beyond their extremal values and compute them for local complete intersections, affine cones, and other explicit singularities. These projects provide natural problems for graduate students and create connections among algebraic geometry, commutative algebra, representation theory, Hodge theory, and K -theory.

REFERENCES

- [1] D. Orlov A. Bondal. Semiorthogonal decomposition for algebraic varieties. [arXiv:9506012](https://arxiv.org/abs/9506012), 1995.
- [2] Rina Anno and Timothy Logvinenko. On adjunctions for Fourier-Mukai transforms. *Adv. Math.*, 231(3-4):2069–2115, 2012.
- [3] Luchezar L. Avramov, Ragnar-Olaf Buchweitz, Srikanth B. Iyengar, and Claudia Miller. Homology of perfect complexes. *Adv. Math.*, 223(5):1731–1781, 2010.
- [4] Matthew R. Ballard, Srikanth B. Iyengar, Pat Lank, Alapan Mukhopadhyay, and Josh Pollitz. High Frobenius pushforwards generate the bounded derived category. *Forum Math. Sigma*, 14:29, 2026. Id/No e12.
- [5] Claudio Bartocci, Ugo Bruzzo, and Daniel Hernández Ruipérez. *Fourier-Mukai and Nahm transforms in geometry and mathematical physics*, volume 276 of *Prog. Math.* Basel: Birkhäuser, 2009.
- [6] Daniel Bergh and Olaf M. Schnürer. Conservative descent for semi-orthogonal decompositions. *Adv. Math.*, 360:39, 2020. Id/No 106882.
- [7] A. Bondal and D. Orlov. Derived categories of coherent sheaves. In *Proceedings of the International Congress of Mathematicians, ICM 2002, Beijing, China, August 20–28, 2002. Vol. II: Invited lectures*, pages 47–56. Beijing: Higher Education Press; Singapore: World Scientific/distributor, 2002.
- [8] A. Bondal and M. Van den Bergh. Generators and representability of functors in commutative and noncommutative geometry. *Mosc. Math. J.*, 3(1):1–36, 2003.
- [9] Tom Bridgeland. Flops and derived categories. *Invent. Math.*, 147(3):613–632, 2002.
- [10] Tom Bridgeland. Stability conditions on triangulated categories. *Ann. Math. (2)*, 166(2):317–345, 2007.
- [11] Ragnar-Olaf Buchweitz. *Maximal Cohen-Macaulay modules and Tate cohomology. With appendices by Luchezar L. Avramov, Benjamin Briggs, Srikanth B. Iyengar and Janina C. Letz*, volume 262 of *Math. Surv. Monogr.* Providence, RI: American Mathematical Society (AMS), 2021.
- [12] Andrew Burke, Mihnea Popa, and Wanchun Shen. Negative k -theory and hodge theory. [arXiv:2607.12672](https://arxiv.org/abs/2607.12672), 2026.
- [13] Timothy De Deyn, Pat Lank, Kabeer Manali Rahul, and Sridhar Venkatesh. Measuring birational derived splinters. *Bull. Lond. Math. Soc.*, 58(5):18, 2026. Id/No e70362.
- [14] Souvik Dey and Pat Lank. Closedness of the singular locus and generation for derived categories. *J. Algebra*, 684:64–77, 2025.
- [15] Souvik Dey and Pat Lank. Dévissage for generation in derived categories. *Proc. Am. Math. Soc.*, 154(3):997–1006, 2026.
- [16] Davis C. Doherty. Singularities of generic projection hypersurfaces. *Proc. Am. Math. Soc.*, 136(7):2407–2415, 2008.
- [17] Uttaran Dutta, Pat Lank, and Kabeer Manali Rahul. Integral transforms on singularity categories for noetherian schemes. [arXiv:2501.13834](https://arxiv.org/abs/2501.13834), 2025. to appear in *Michigan Math. J.*
- [18] Robert Friedman and Radu Laza. Higher Du Bois and higher rational singularities. *Duke Math. J.*, 173(10):1839–1881, 2024.
- [19] Elías Guisado Villalgorido and Pat Lank. Openness of fourier-mukai loci for algebraic spaces. Available at <https://lankp.github.io/assets/PDFs/OpenFMLocus.pdf>, 2026.
- [20] Elías Guisado Villalgorido, Pat Lank, Kabeer Manali Rahul, and Nebojsa Pavic. Yoga for Fourier–Mukai partnership. [arXiv:2512.16503](https://arxiv.org/abs/2512.16503), 2025.
- [21] Jack Hall and Kyle Priver. A generalized Bondal-Orlov full faithfulness criterion for Deligne-Mumford stacks. *Sel. Math., New Ser.*, 32(3):47, 2026. Id/No 51.
- [22] Seung-Jo Jung, In-Kyun Kim, Morihiro Saito, and Youngho Yoon. Higher du Bois singularities of hypersurfaces. *Proc. Lond. Math. Soc. (3)*, 125(3):543–567, 2022.
- [23] Martin Kalck. A new equivalence between singularity categories of commutative algebras. *Adv. Math.*, 390:11, 2021. Id/No 107913.
- [24] Yujiro Kawamata. D -equivalence and K -equivalence. *J. Differ. Geom.*, 61(1):147–171, 2002.
- [25] Horst Knörrer. Cohen-Macaulay modules on hypersurface singularities. I. *Invent. Math.*, 88:153–164, 1987.
- [26] János Kollár. Simultaneous normalization and algebra husks. *Asian J. Math.*, 15(3):437–450, 2011.
- [27] Sándor Kovács, Pat Lank, and Sridhar Venkatesh. Simple criteria for higher rational singularities. [arXiv: 2507.07351](https://arxiv.org/abs/2507.07351), 2025.
- [28] Sándor J. Kovács. A characterization of rational singularities. *Duke Math. J.*, 102(2):187–191, 2000.
- [29] Sándor J. Kovács. Complexes of differential forms and singularities: The injectivity theorem. [arXiv:2505.09912](https://arxiv.org/abs/2505.09912), 2025.
- [30] Pat Lank. Descent conditions for generation in derived categories. *J. Pure Appl. Algebra*, 228(9):19, 2024. Id/No 107671.
- [31] Pat Lank, Peter M. McDonald, and Sridhar Venkatesh. Measuring rationality of Schwede-Takagi pairs. *J. Pure Appl. Algebra*, 230(6):9, 2026. Id/No 108268.
- [32] Pat Lank and Noah Olander. Approximation by perfect complexes detects Rouquier dimension. *Mosc. Math. J.*, 25(1):13–31, 2025.
- [33] Pat Lank and Fei Peng. Frobenius generation for algebraic stacks. [arXiv:2512.05026](https://arxiv.org/abs/2512.05026), 2025.
- [34] Pat Lank and Sridhar Venkatesh. Triangulated characterizations of singularities. *Nagoya Math. J.*, 261:15, 2026. Id/No e3.
- [35] Hiroki Matsui and Ryo Takahashi. Singularity categories and singular equivalences for resolving subcategories. *Math. Z.*, 285(1-2):251–286, 2017.
- [36] Shigeru Mukai. Duality between $D(X)$ and $D(\hat{X})$ with its application to Picard sheaves. *Nagoya Math. J.*, 81:153–175, 1981.
- [37] Mircea Mustață, Sebastián Olano, Mihnea Popa, and Jakub Witaszek. The Du Bois complex of a hypersurface and the minimal exponent. *Duke Math. J.*, 172(7):1411–1436, 2023.
- [38] Mircea Mustață and Mihnea Popa. Hodge filtration on local cohomology, Du Bois complex and local cohomological dimension. *Forum Math. Pi*, 10:58, 2022. Id/No e22.
- [39] D. O. Orlov. Derived categories of coherent sheaves on abelian varieties and equivalences between them. *Izv. Math.*, 66(3):569–594, 2002.
- [40] D. O. Orlov. Triangulated categories of singularities and D-branes in Landau-Ginzburg models. In *Algebraic geometry. Methods, relations, and applications. Collected papers. Dedicated to the memory of Andrei Nikolaevich Tyurin.*, pages 227–248. Moscow: Maik Nauka/Interperiodica, 2004.
- [41] Nebojsa Pavic and Evgeny Shinder. K -theory and the singularity category of quotient singularities. *Ann. K-Theory*, 6(3):381–424, 2021.
- [42] Morihiro Saito. Introduction to mixed Hodge modules. In *Actes du colloque de théorie de Hodge. Luminy, France, 1-6 Juin 1987*, pages 145–162. Paris: Société Mathématique de France, 1989.
- [43] Morihiro Saito. Mixed Hodge modules. *Publ. Res. Inst. Math. Sci.*, 26(2):221–333, 1990.
- [44] Karl Schwede. A simple characterization of Du Bois singularities. *Compos. Math.*, 143(4):813–828, 2007.
- [45] Wanchun Shen. On higher du bois singularities and K -regularity. [arXiv:2504.12402](https://arxiv.org/abs/2504.12402), 2025.
- [46] Wanchun Shen, Sridhar Venkatesh, and Anh Duc Vo. On k -Du Bois and k -rational singularities. [arXiv:2306.03977](https://arxiv.org/abs/2306.03977), 2023.
- [47] R. W. Thomason. The classification of triangulated subcategories. *Compos. Math.*, 105(1):1–27, 1997.
- [48] Anh Duc Vo. *Du Bois complexes and singularity theory*. Ph.D. Dissertation, Harvard University, Cambridge, MA, 2025. Department of Mathematics.

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